3.3 HUFFMANOV KOD

FAZE:

1) ZDRUŽEVANJE

a) ZDRUŽIMO k NAJMANJ VERJETNIH ZNAKOV V 1 ZNAK

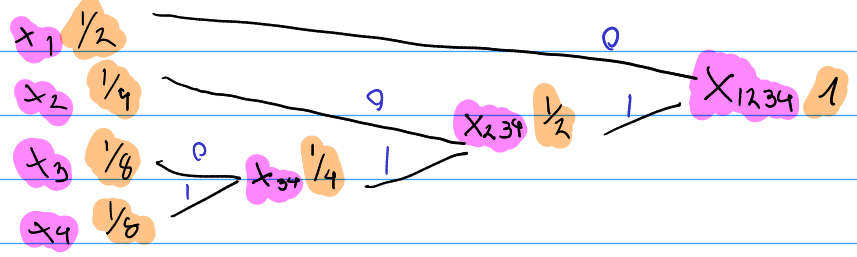
p ZNAKA $= \sum p_i$

b) REKURZIVNO PONAVLJAMO

2) RAZDRUŽEVANJE

- VSAKEMU ZNAKU IZ SKUPNEGA ZNAKA DOBI SVOJ ZNAK K.ABC.

PRIMER 1)

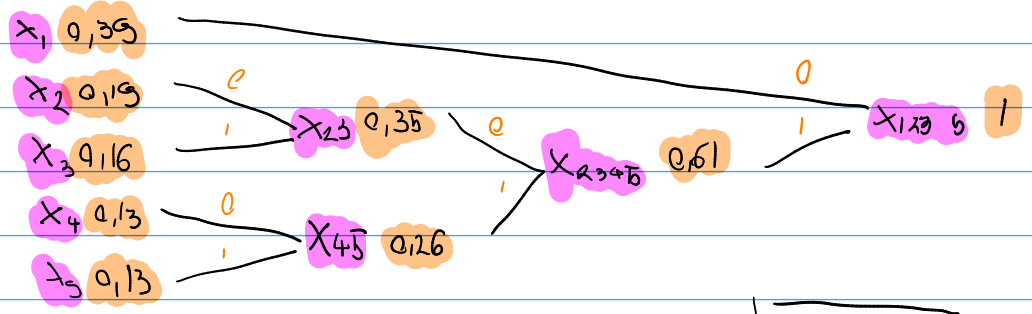


x_1	0	x_3	110
x_2	10	x_4	111

$H = L = \frac{7}{4}$

$\eta = 1$

PRIMER 2)



x_1	0	x_3	101	x_5	111
x_2	100	x_4	110		

$L = 2{,}22$

$H = 2{,}17$

$\eta = 0{,}98$

ALI JE HUFFMAN RES OPTIMALN?

- NIČ NEIZRABLJENIH VEJ
- NAJVEČJI SO V PARIH
- NAJVEČJE ZAMENJAVE → NAJMANJ VRJETNI ZNAKI

VSAK OPTIMALNI KOD NI HUFFMANOV

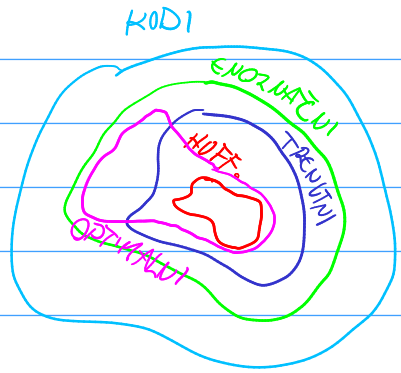
PRIMER:

$P = \{1/2, 1/4, 1/8, 1/8\}$

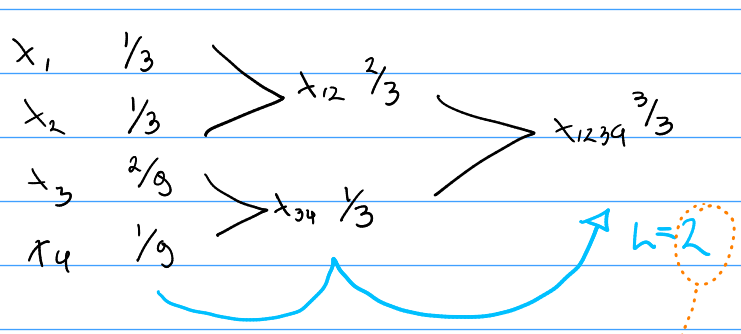
$C_H = \{0, 10, 110, 111\}$ → HUFFMANOV

$C_? = \{0, 01, 010, 111\}$

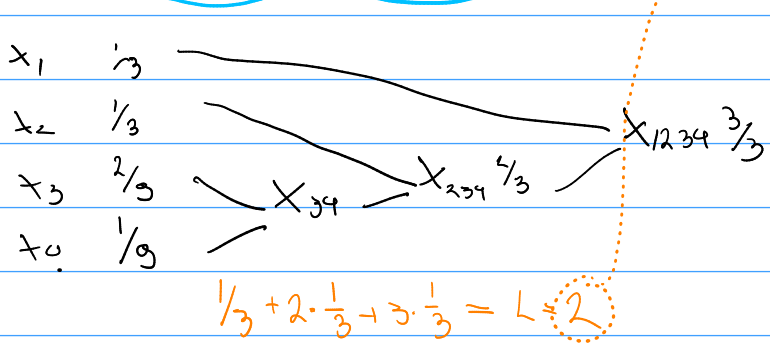
NI ISTI, PA JE LEGIT



HUFF 1



HUFF 2



HUFFMAN $r > 2$ TUD OBSTAJA

- ČE SE NE IZIDE DODAMO PRAZEN ZNAK X_0 i $p=0$

n = ZNAKOV

$n = k(r-) + r$

$\boxed{n-1 = k'(r-1) \quad , k' \in N}$

PRIMER: a, b $\quad p_a = 0{,}0001 \quad p_b = 0{,}9999$

LAHKO KODIRAMO BLOKE $\quad X = \{a, b\} \quad P = \{0{,}75, 0{,}25\}$

a	0	aa	$9/16$	0
b	1	ab	$3/16$	0
		ba	$3/16$	0
		bb	$1/16$	1

$\frac{4}{16}$ 1 $\quad \frac{7}{16}$ 1 $\quad$ 1

$L = 0{,}844$

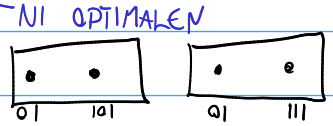
VEČ BLOKOV → BOLJŠI L

a	$3/4$	1
b	$1/4$	

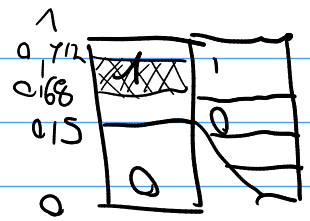
$L = 1$

ARITMETIČNI KOD

- # ITER
- NI OPTIMALEN



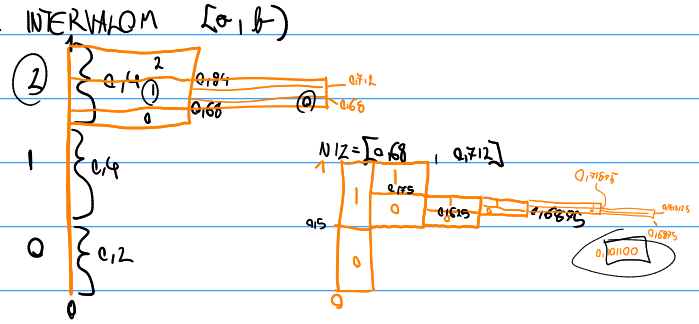
PRIMER 2



- NE RAZVRŠČAMO
- INTERVAL $(0, 1)$
  - ↳ RAZBITJE NA n INTERVALOV
  - ↳ ŠIRINA USTREZA VRJETNOSTI ZNAKOV
- IZBEREMO USTREZEN PODINTERVAL.
- ČE NIZ NI KONČAN

- NIZ PREDSTAVIMO Z INTERVALOM $[a, b)$

$A = \{0, 1, 2\}$

$P = \{0{,}2, 0{,}4, 0{,}4\}$

$B = \{0, 1\}$